From Latent Structure to Predictive Modeling: A Statistical Investigation of Abalone Morphology

Hexiang Li
Department of Statistical Science
University of Toronto

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1 Introduction

This dataset consists of physical measurements of abalone collected from the coast of Tasmania, Australia. Each observation records morphological characteristics such as length, diameter, height, and various weight measurements. Traditionally, the age of an abalone is determined by counting the number of rings on its shell, a process that is time-consuming, labor-intensive, and requires specialized expertise. As a result, physical measurements are often used as proxies for age, enabling scalable and non-destructive analysis.

The dataset is obtained from the UCI Machine Learning Repository and is one of the oldest and most widely used benchmark datasets in the repository. Its longevity and continued use reflect both the richness of its multivariate structure and its relevance for studying relationships among correlated measurements.

From a statistical perspective, this dataset is particularly interesting because it contains multiple highly correlated variables that describe similar aspects of abalone morphology. Such redundancy makes the data well-suited for dimensionality reduction techniques such as Principal Component Analysis (PCA). In this project, we use PCA to explore whether the high-dimensional measurement space admits a meaningful low-dimensional structure and to assess the stability and significance of this structure using resampling-based inference.

2 Exploratory Data Analysis

2.1 Summary Statistics

The dataset contains 4,177 complete observations for each of the seven continuous variables. Clear differences in scale are observed across measurements. Linear size variables such as length and diameter have moderate dispersion (means around 0.52 and 0.41), whereas weight-related variables exhibit substantially larger variability. In particular, whole weight shows the greatest spread, ranging from near zero to approximately 2.83, indicating the presence of extreme values.

Height displays comparatively low variability, with a mean of 0.14 and a small standard deviation, suggesting a more concentrated distribution. The interquartile ranges further confirm that weight variables vary more widely than size measurements. These differences in magnitude motivate standardization prior to conducting PCA, ensuring that larger-scale variables do not dominate the analysis.

2.2 Boxplots

Figure 1 presents boxplots of the seven continuous input variables, representing different physical dimensions and weight components of abalone. The boxplots reveal pronounced differences among variables, particularly in terms of median values and overall spread. With the exception of height, most variables exhibit relatively large interquartile ranges, indicating substantial variability across observations. Among these, whole weight shows the greatest dispersion, likely reflecting the combined variability of shell weight, viscera weight, and shucked weight.

Most variables display noticeable skewness, with weight-related measurements exhibiting long right tails and numerous upper outliers, consistent with right-skewed distributions. In contrast, length and diameter show several lower-end outliers, suggesting mild left skewness. Height appears more tightly concentrated for the majority of observations, though a small number of extreme values are present.

Overall, the boxplots highlight heterogeneity in both scale and distributional shape, motivating standardization before applying PCA.

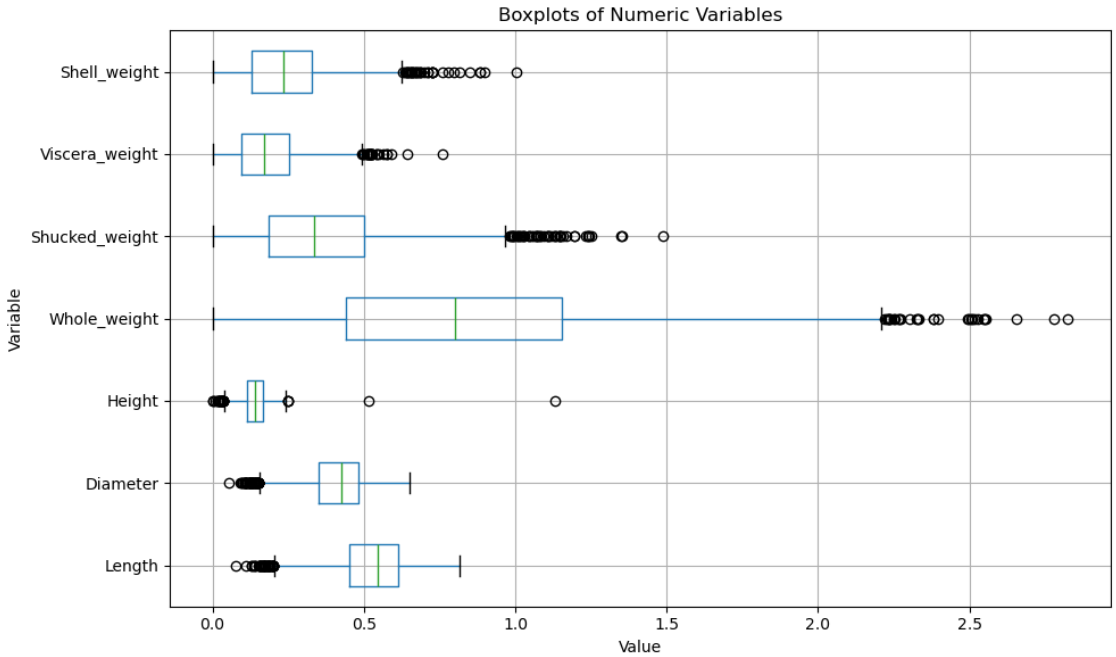

Figure 1: Boxplots of the seven continuous morphometric variables, illustrating differences in scale, dispersion, skewness, and the presence of outliers across measurements.

2.3 Correlation Structure

The four heatmaps provide a detailed view of the correlation structure across the full sample and within each sex category. Across all observations, most pairwise correlations fall between 0.80 and 1.00, indicating substantial redundancy among the variables. In particular, both size-related and weight-related measurements exhibit consistently strong positive associations. This pervasive dependence suggests that the observed measurements likely reflect a common underlying latent size factor, motivating the use of Principal Component Analysis (PCA) to investigate the intrinsic dimensionality of the data.

Comparing the four panels reveals that strong positive correlations persist within each subgroup. For males, correlations generally range from approximately 0.80 to 1.00. For females, correlations remain high overall but are somewhat more variable, with several associations in the 0.50–0.60 range, particularly involving height. In contrast, the infant group exhibits the strongest and most uniform dependence structure, with correlations predominantly between 0.90 and 1.00.

Importantly, the direction and magnitude of correlations remain broadly consistent across all groups. This stability indicates that the strong positive associations observed in the pooled data are not driven by aggregation artifacts. In other words, there is no evidence that the

overall correlation structure masks substantially different subgroup patterns.

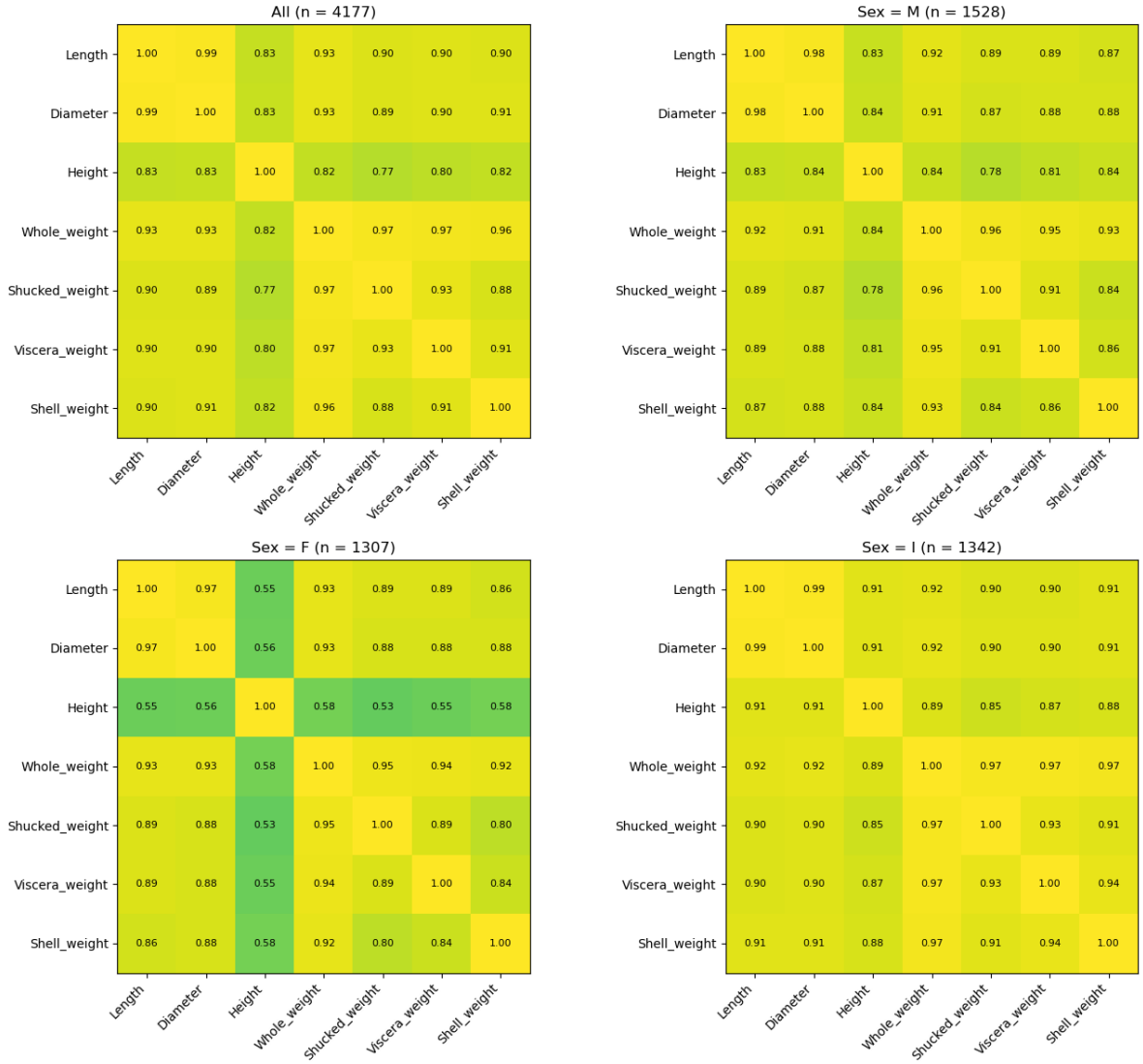

Figure 2: Pearson correlation matrices for all observations and stratified by sex (Male, Female, Infant), illustrating strong positive associations and structural similarity across groups.

2.4 Principal Component Analysis

2.4.1 Results

Motivated by the strong redundancy observed in the correlation structure, we apply Principal Component Analysis (PCA) to investigate the intrinsic dimensionality of the measurement space.

The scree plot is used for component selection due to its intuitive visualization of explained

variance across components. As shown in Figure 3, the first principal component (PC1) explains nearly all of the total variance, while the remaining components contribute only negligible proportions. The sharp drop after the first component and the near-flat pattern thereafter indicate a dominant one-dimensional structure in the data. This suggests that most of the variability in the seven measurements can be summarized by a single latent factor.

Nevertheless, for interpretability and comparison, we retain the first two principal components in subsequent visualization.

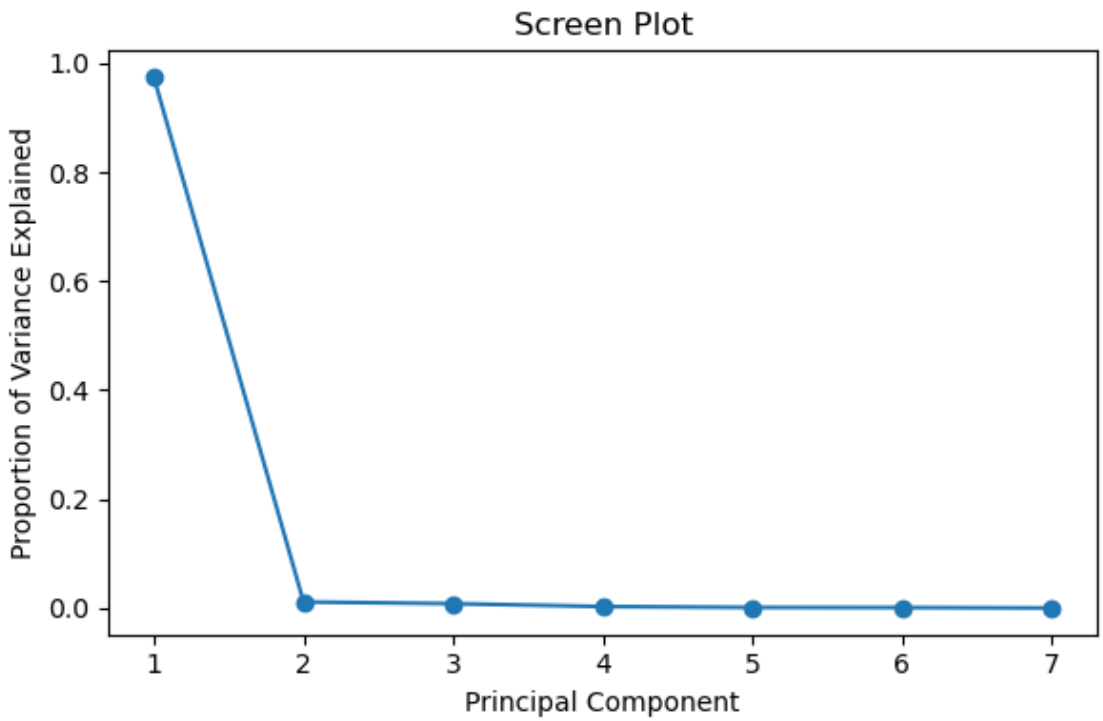

Figure 3: Scree plot of the principal components showing the proportion of variance explained, indicating a dominant first component and rapid decay thereafter.

Subsequently, the biplot displays both the scores of observations and the loadings of the original variables on the first two principal components. Since PC1 explains almost all of the variance, it can be interpreted as an overall "size" or "mass" component. All size and weight-related variables load strongly and positively on PC1, indicating that they increase together. This confirms the earlier observation of pervasive positive correlations.

In contrast, PC2 captures a much smaller proportion of variance and reflects secondary variation orthogonal to overall size. The loading vector for height shows a relatively distinct direction compared to the weight variables, suggesting that PC2 may represent shape-related

variation rather than pure size. The loadings indicate that PC2 contrasts height with several mass-related variables, suggesting subtle differences in body shape rather than overall size. This component may reflect variation in proportional growth rather than total mass.

The mass-related variables (whole weight, shucked weight, viscera weight, and shell weight) have very small angles between their loading vectors, indicating strong positive correlations and substantial redundancy. Their nearly aligned directions visually confirm the clustering observed in the heatmaps (Figure 2).

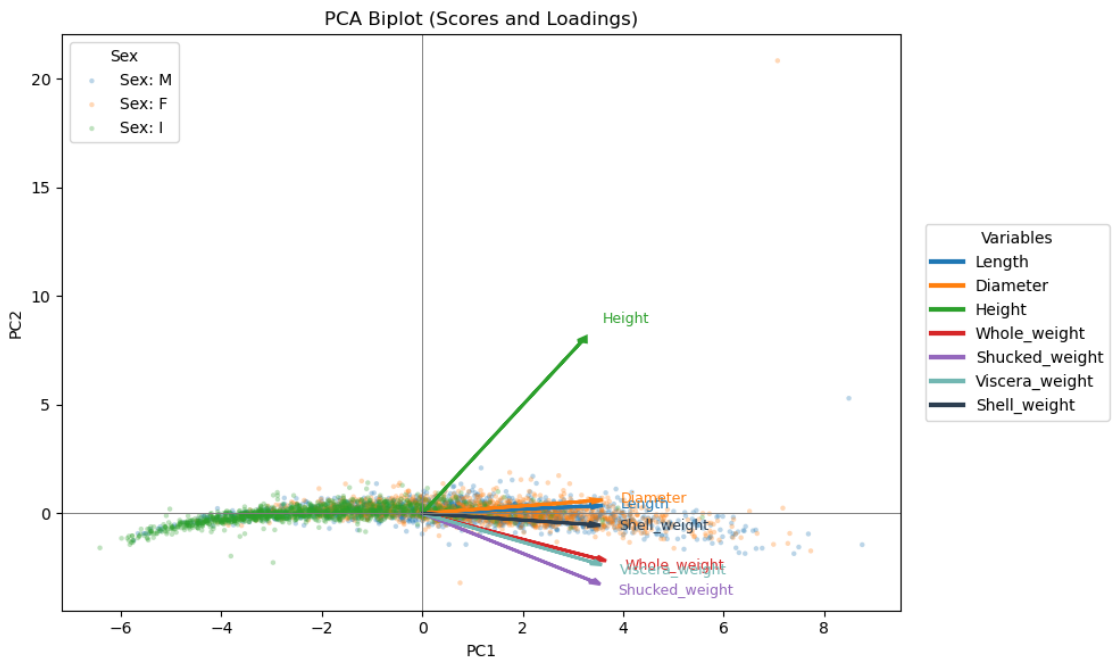

Figure 4: PCA biplot of PC1 and PC2 showing observation scores by sex and loading vectors of original variables, highlighting the dominant common size/mass factor underlying the measurements.

Overall, the PCA results provide strong evidence that abalone morphology is dominated by a single underlying latent dimension corresponding to overall body size, with only minor secondary variation. This suggests that abalone growth is largely proportional across physical dimensions, with larger individuals tending to have proportionally greater shell, viscera, and meat weights, consistent with a biological scaling process.

2.4.2 Inferences on Principal Component

To formally evaluate the significance of the principal components, we conducted a permutation test under the null hypothesis of no cross-variable correlation. Under this null dis-

tribution, only the first principal component (PC1) was found to be statistically significant. Specifically, the observed eigenvalue for PC1 was 6.36, with a permutation-based p-value smaller than 0.001. In contrast, all remaining components (PC2–PC7) had p-values equal to 1.00, indicating that their observed explained variance ratios are entirely consistent with what would be expected under random independence and pre-specified $\alpha = 0.05$.

In terms of variance explained, PC1 accounts for approximately 90.8 percent of the total variance (EVR = 0.9079), while PC2 explains only about 4.0 percent, and each subsequent component contributes less than 2.5 percent. This sharp drop reinforces the dominance of PC1.

To assess stability, we performed bootstrap resampling. The bootstrap 95 percent confidence interval for the explained variance ratio of PC1 ranges from approximately 0.874 to 0.927. This interval remains far above the contributions of all other components, confirming that the dominance of PC1 is highly stable under sampling variability. In contrast, PC2 has a bootstrap confidence interval of roughly 0.029 to 0.073, demonstrating both its small magnitude and limited structural importance.

Taken together, the permutation and bootstrap results provide strong evidence that the abalone morphology measurements are governed primarily by a single latent dimension. PC1 is not only statistically significant but also robust, whereas the remaining components do not represent meaningful additional structure. Without such inference procedures, the dominance of PC1 could potentially reflect sample-specific structure; the permutation and bootstrap results demonstrate that the observed low-dimensional pattern is not an artifact of random variation.

2.5 Discussion

This analysis reveals that abalone morphological measurements are highly redundant. The correlation heatmaps show strong positive relationships among size and weight variables, suggesting the presence of an underlying low-dimensional structure. PCA confirms this: the first principal component alone explains about 91 percent of the total variance. Both permutation testing and bootstrap resampling indicate that this dominant component is statistically significant and stable.

The first principal component can be interpreted as an overall body size factor, as all length and weight variables load strongly in the same direction. Additional components explain only small proportions of variance and likely reflect minor shape differences rather than

distinct structural dimensions.

However, PCA captures only linear relationships and is unsupervised, meaning that components explaining high variance may not be optimal for predicting age (ring count). The dominance of a single latent size dimension suggests that abalone growth is largely governed by proportional biological scaling, rather than independent variation across physical components. This indicates that most morphological measurements increase together as overall body size increases. Future work could incorporate supervised models, such as Poisson regression, to assess how this underlying morphological structure relates to growth and age.

3 Predictive Modeling of Abalone Age

3.1 Modeling Objective and Framework

The objective of this section is to construct statistically principled predictive models for *Rings*, a non-negative integer response variable representing abalone age.

Given the discrete count nature of the outcome, generalized linear models (GLMs) provide a natural modeling framework. Specifically, we consider models of the form

$$\log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \tag{1}$$

where

$$\mu_i = \mathbb{E}[Y_i \mid \mathbf{x}_i]$$

and Y_i denotes the ring count.

Two distributional assumptions are considered:

- **Poisson GLM** (equidispersion assumption)
- **Negative Binomial GLM** (allows overdispersion)

To ensure unbiased assessment of predictive performance, the dataset is randomly partitioned into:

- 80% training set (estimation)
- 20% testing set (evaluation)

All model comparisons are conducted on identical splits to maintain fairness.

3.2 Distributional Diagnostics and Overdispersion Assessment

A key assumption of the Poisson model is

$$\mathrm{Var}(Y_i) = \mathbb{E}[Y_i]. \tag{2}$$

However, empirical inspection of the training set yields a variance-to-mean ratio of approximately 1.0388, indicating mild overdispersion.

Although the deviation from unity is small, even moderate overdispersion can:

- Underestimate standard errors
- Inflate Type I error rates
- Bias model comparison metrics

To safeguard inference and ensure robustness, we adopt the **Negative Binomial GLM with log link** throughout.

Under this specification,

$$\mathrm{Var}(Y_i) = \mu_i + \alpha\mu_i^2, \tag{3}$$

where α is estimated via maximum likelihood.

This formulation provides flexibility while retaining interpretability of log-linear effects.

3.3 Model Selections

3.4 Baseline Model: Full Morphological Specification

We first estimate a Negative Binomial GLM including all seven morphological predictors (Table 1).

Table 1: Negative Binomial GLM Results (Original Morphological Variables)

Model Information				
Dependent Variable	Rings			
Model Family	Negative Binomial (log link)			
Deviance	1494.5			
Pseudo R^2 (Cox–Snell)	0.4152			
AIC	15218.046			
Coefficient Estimates				
Variable	Coef.	Std. Err.	z-value	p-value
Intercept	2.2694	0.006	402.773	$< 0.001^{***}$
Length	0.0158	0.034	0.461	0.645
Diameter	0.1801	0.034	5.224	$< 0.001^{***}$
Height	0.0408	0.008	5.152	$< 0.001^{***}$
Whole_weight	0.3824	0.052	7.365	$< 0.001^{***}$
Shucked_weight	-0.4200	0.027	-15.419	$< 0.001^{***}$
Viscera_weight	-0.0850	0.021	-4.048	$< 0.001^{***}$
Shell_weight	0.0705	0.022	3.153	0.002^{**}

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Key Findings

- Nearly all predictors are statistically significant.
- The model achieves low deviance and high pseudo-R^2 among all specifications.

However, earlier PCA and correlation analysis revealed substantial redundancy among predictors, raising concerns regarding multicollinearity.

Importantly, multicollinearity does not necessarily degrade predictive accuracy, but it does compromise interpretability and coefficient stability.

Thus, although the full model performs best predictively, further structural investigation is warranted.

3.5 PCA-Based Dimensionality Reduction Model

Motivated by Section 2, we estimate a reduced model using PC1 and PC2 as predictors (Table 2).

Table 2: Negative Binomial GLM Results (PC1 and PC2)

Model Information				
Dependent Variable	Rings			
Model Family	Negative Binomial (log link)			
Deviance	2097.1			
Pseudo R^2 (Cox–Snell)	0.2996			
AIC	15810.670			
Coefficient Estimates				
Variable	Coef.	Std. Err.	z-value	p-value
Intercept	2.2787	0.006	408.183	$< 0.001^{***}$
PC1	0.0729	0.002	33.305	$< 0.001^{***}$
PC2	0.0588	0.006	10.255	$< 0.001^{***}$

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Interpretation

- Both PC1 and PC2 are highly statistically significant.
- Deviance and AIC increase substantially relative to the full model.
- Pseudo R^2 decreases.

This illustrates an important modeling insight: variance-explaining directions obtained from unsupervised PCA are not guaranteed to be optimal for supervised prediction.

PCA maximizes total variance in predictors, not covariance with the response variable *Rings*.

Therefore, despite strong latent structure, PCA-based regression sacrifices predictive power.

3.6 Multicollinearity Diagnostics

Variance Inflation Factors (VIFs) confirm severe multicollinearity:

- Whole_weight: VIF > 100
- Length and Diameter: VIF > 40

Such magnitudes suggest:

- Inflated standard errors
- Instability under small perturbations
- Difficulty interpreting marginal effects

Given the biological interpretability of predictors, structural refinement is preferable to blind regularization.

3.7 Structurally Reduced Models

We explore biologically motivated alternatives.

(1) Reduced Weight Model

Retain only **Whole_weight**.

Result

- VIF decreases substantially.
- Predictive performance deteriorates moderately.

(2) Area Model

Replace Length and Diameter with

$$\text{Area} = \text{Length} \times \text{Diameter}.$$

Result

- Interpretability improves biologically.
- Predictive performance declines further.

(3) No-Length Model

Retain Diameter, Height, and Whole_weight.

Result

- VIFs fall below conventional thresholds (< 10).
- Performance is only slightly worse than the full model.
- Coefficients remain stable and interpretable.

Among reduced models, the No-Length specification achieves the most favorable trade-off.

3.8 Out-of-Sample Evaluation

All model specifications are evaluated using the following out-of-sample metrics:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Test Deviance

Results (Table 3) indicate:

Table 3: Out-of-Sample Performance Comparison of Negative Binomial Models

Model	RMSE	MAE	Test Deviance
NB–GLM Original (7 vars)	2.334	1.672	402.635
NB–GLM Reduced (drop 3 weights)	2.659	1.931	521.291
NB–GLM No Length (Diameter + Height + W)	2.665	1.939	523.126
NB–GLM PCA (PC1 + PC2)	2.687	1.951	531.360
NB–GLM Area (Area + Height + W)	2.713	1.982	547.817

- The full model achieves the lowest RMSE and deviance.
- Reduced models incur moderate predictive cost.

- The PCA model performs worst among candidates.

These findings confirm that retaining detailed morphological information improves predictive generalization.

3.9 Final Model Selection: Accuracy vs. Interpretability

Two competing modeling objectives emerge.

If the goal is predictive accuracy:

The full seven-variable Negative Binomial model is preferred.

If the goal is interpretability and structural clarity:

The No-Length model provides a stable and biologically coherent alternative.

This illustrates a fundamental statistical principle:

> *Model selection is objective-dependent; no single model dominates across all criteria.*

The final recommendation depends on whether predictive performance or interpretability is prioritized.

Appendix A: Full Regression Outputs

Table 4: Negative Binomial GLM Results (PC1 and PC2)

Model Information				
Dependent Variable	Rings			
Model Family	Negative Binomial (log link)			
Link Function	Log			
Deviance	2097.1			
Pseudo R^2 (Cox–Snell)	0.2996			
AIC	15810.670332			
Coefficient Estimates				
Variable	Coef.	Std. Err.	z-value	p-value
Intercept	2.2787	0.006	408.183	$< 0.001^{***}$
PC1	0.0729	0.002	33.305	$< 0.001^{***}$
PC2	0.0588	0.006	10.255	$< 0.001^{***}$

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Table 5: Negative Binomial GLM Results (Reduced Model)

Model Information				
Dependent Variable	Rings			
Model Family	Negative Binomial (log link)			
Deviance	2027.2			
Pseudo R^2 (Cox–Snell)	0.3141			
Coefficient Estimates				
Variable	Coef.	Std. Err.	z-value	p-value
Intercept	2.2764	0.006	406.409	$< 0.001^{***}$
Length	-0.0992	0.034	-2.955	0.003^{**}
Diameter	0.2818	0.034	8.375	$< 0.001^{***}$
Height	0.0545	0.007	8.314	$< 0.001^{***}$
Whole_weight	-0.0314	0.015	-2.114	0.035^{*}

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Table 6: Negative Binomial GLM Results (Area Model)

Model Information				
Dependent Variable	Rings			
Model Family	Negative Binomial (log link)			
Deviance	2146.7			
Pseudo R^2 (Cox–Snell)	0.2891			
Coefficient Estimates				
Variable	Coef.	Std. Err.	z-value	p-value
Intercept	2.2791	0.006	408.285	$< 0.001^{***}$
Area	0.1310	0.020	6.715	$< 0.001^{***}$
Height	0.0651	0.006	11.104	$< 0.001^{***}$
Whole_weight	-0.0028	0.019	-0.149	0.881

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Table 7: Negative Binomial GLM Results (No Length Model)

Model Information				
Dependent Variable	Rings			
Model Family	Negative Binomial (log link)			
Deviance	2035.8			
Pseudo R^2 (Cox–Snell)	0.3123			
Coefficient Estimates				
Variable	Coef.	Std. Err.	z-value	p-value
Intercept	2.2765	0.006	406.369	$< 0.001^{***}$
Diameter	0.1939	0.016	12.304	$< 0.001^{***}$
Height	0.0547	0.007	8.354	$< 0.001^{***}$
Whole_weight	-0.0411	0.015	-2.828	0.005^{**}

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Table 8: Negative Binomial GLM Results (Original Morphological Variables)

Model Information				
Dependent Variable	Rings			
Model Family	Negative Binomial (log link)			
Deviance	1494.5			
Pseudo R^2 (Cox–Snell)	0.4152			
AIC	15218.045982			
Coefficient Estimates				
Variable	Coef.	Std. Err.	z-value	p-value
Intercept	2.2694	0.006	402.773	$< 0.001^{***}$
Length	0.0158	0.034	0.461	0.645
Diameter	0.1801	0.034	5.224	$< 0.001^{***}$
Height	0.0408	0.008	5.152	$< 0.001^{***}$
Whole_weight	0.3824	0.052	7.365	$< 0.001^{***}$
Shucked_weight	-0.4200	0.027	-15.419	$< 0.001^{***}$
Viscera_weight	-0.0850	0.021	-4.048	$< 0.001^{***}$
Shell_weight	0.0705	0.022	3.153	0.002^{**}

$^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

Appendix B: Variance Inflation Factors

Variance Inflation Factors are computed for the baseline of Negative Binomial model using the standard OLS auxiliary regression definition and its conventional threshold is VIF < 10:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}$$

Table 9: Variance Inflation Factors: Area Model

Variable	VIF
Area	13.718
Height	3.045
Whole_weight	13.249

Table 10: Variance Inflation Factors: Original Model

Variable	VIF
Length	40.274
Diameter	41.665
Height	3.345
Whole_weight	109.196
Shucked_weight	28.449
Viscera_weight	17.043
Shell_weight	21.123

Table 11: Variance Inflation Factors: Reduced model

Variable	VIF
Length	39.285
Diameter	40.457
Height	3.257
Whole_weight	7.560

Table 12: Variance Inflation Factors: No-Length Model

Variable	VIF
Diameter	3.2
Height	2.4
Whole_weight	3.8

The VIF results reveal severe multicollinearity in the original morphological specification. In particular, Whole_weight exhibits an extremely high VIF (109.196), while Length and Diameter both exceed 40, indicating substantial linear dependence among size-related predictors. Weight-related components also display elevated VIF values, confirming the redundancy suggested by earlier correlation and PCA analysis.

The Reduced model and Area model partially mitigates multicollinearity through variable elimination and interaction. The No-Length specification achieves the most substantial improvement. All VIF values fall well below conventional thresholds (VIF < 10), with the

maximum equal to 3.8. This confirms that removing Length effectively resolves the dominant source of collinearity while preserving predictive structure.

Overall, the VIF diagnostics support the conclusion that the No-Length model provides the most stable and interpretable specification among the reduced alternatives.

Appendix C: PCA Inference Outputs

Table 13: Permutation Test for PCA Eigenvalues and Explained Variance Ratios

PC	Observed Eigenvalue	p-value (Eigenvalue)	Observed EVR	p-value (EVR)
PC1	6.3566	0.0010	0.9079	0.0010
PC2	0.2795	1.0000	0.0399	1.0000
PC3	0.1674	1.0000	0.0239	1.0000
PC4	0.1141	1.0000	0.0163	1.0000
PC5	0.0647	1.0000	0.0092	1.0000
PC6	0.0127	1.0000	0.0018	1.0000
PC7	0.0067	1.0000	0.0010	1.0000

EVR = Explained Variance Ratio. Permutation-based p-values are computed using randomized covariance structure.

Table 14: Bootstrap Confidence Intervals for Explained Variance Ratios

PC	Observed EVR	95% CI (Lower)	95% CI (Upper)
PC1	0.9079	0.8743	0.9267
PC2	0.0399	0.0289	0.0728
PC3	0.0239	0.0189	0.0253
PC4	0.0163	0.0125	0.0175
PC5	0.0092	0.0085	0.0099
PC6	0.0018	0.0016	0.0021
PC7	0.0010	0.0008	0.0011

The permutation results indicate that only the first principal component is statistically significant ($p = 0.001$), while all subsequent components are indistinguishable from random

noise. Bootstrap confidence intervals further confirm that PC1 explains a dominant proportion of total variance, with a narrow and stable 95% interval. These findings justify retaining PC1 as the primary latent morphological dimension.

References

James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An Introduction to Statistical Learning: With Applications in R*. 2nd ed., Springer, 2021.

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